One commenter suggested that the California laws could increase compliance costs by increasing demand, and thus the cost, for external consultants and services.[3118] We acknowledge this could occur in the short term; however, over the long-term, we expect that increased demand would cause new providers to enter the market, resulting in a corresponding increase in supply. An increase in the supply of providers would lead to greater competition among the external consultants, resulting in lower fees charged by consultants. To that end, the phased implementation of the final rules should mitigate most costs stemming from any shortage of consultants.

Registrants that have more exposure to material climate-related risks may face higher compliance costs to the extent that they must provide more extensive disclosures. However, we note that industries in which climate-risks are most likely to be material are also those that are already providing some degree of voluntary or mandatory disclosures.[3119]

The incremental costs of the financial statement disclosures may be somewhat higher for companies with exposure to severe weather events or other natural conditions that are difficult to assess, track, and disclose in the financial statements. For example, companies (e.g., banks) with complicated asset structures or with operations in many jurisdictions may incur more costs to identify the expenditures for which a severe weather event or other natural condition was a "significant contributing factor."

Incremental costs, either proportionally or in dollar terms, may be higher for smaller registrants, such as SRCs and EGCs, considering that they are less likely to have climate-related

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[3118] *See* letter from Chamber II; *see also supra* note 3125 and accompanying text.

[3119] *See* section IV.A.5.a.

disclosure systems and processes already in place.[3120] If smaller firms were to face higher proportional fixed costs in meeting the disclosure requirements, they may potentially be placed at a competitive disadvantage relative to larger firms.[3121] Conversely, incremental costs may be lower for smaller firms to the extent that their assets and operations are less complex, which may allow them to prepare responsive disclosures at lower cost. We recognize that a portion of the final rules' compliance costs is "fixed" in the sense that the costs do not scale with registrant size or its level of resources. We therefore expect that smaller registrants will have more difficulty allocating resources to comply with the final rules as compared to larger firms.[3122] To mitigate these compliance burdens, the final rules provide SRCs and EGCs certain accommodations, including being exempt from the GHG emissions disclosure requirement and the accompanying assurance requirement, as well as an extended phased in compliance period, which will allow such issuers both more time to prepare for initial compliance, as well as the benefit of observing market practices prior to preparing their initial disclosures required in response to the final rules.

We expect compliance costs to decrease over time. For example, a registrant disclosing climate-related information for the first time is likely to incur initial fixed costs to develop and implement the necessary processes and controls.[3123] Once the company invests in the

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[3120] Commission staff's analysis of registrants' annual filings indicate that SRCs and EGCs are less likely to have climate-related disclosures (as indicated by the presence of climate-related keywords) within their filings (*see* section IV.A.5.a); *see also* section IV.A.5.b.ii for another Commission staff analysis that finds that SRCs and EGCs are less likely to disclose GHG emissions.

[3121] *See, e.g.*, letters from Chamber and NAM.

[3122] *See, e.g.*, letter from CrowdCheck Law ("For example, for two companies we have worked with that recently became Exchange Act reporting companies, the estimated costs for the first year of compliance with the proposed rules would represent approximately 18.5% and 15%, respectively, of their entire gross revenues for the year prior to becoming a reporting company."); *see also* letter from Independent Community Bankers (stating that "the compliance cost burden for the smallest community banks is double that of the largest community banks").

[3123] *See* letter from Financial Executives International's ("FEI") Committee on Corporate Reporting ("CCR") (June 10, 2021); *see also* Proposing Release section IV.C.4.c.

institutional knowledge and systems to prepare the disclosures, the procedural efficiency of these processes and controls should subsequently improve, leading to lower costs in subsequent years.

Mandated climate disclosures may heighten demand for third-party services related to preparing the required disclosures, especially if registrants' current service providers cannot provide the specific services that registrants may seek to comply with the final rules.[3124] In the short term, there could be a potential increase in the prices of such services, leading to higher compliance costs. In the long term, however, this heightened demand is expected to spur competition, innovation, and economies of scale that could over time lower associated costs for such services and improve their availability.[3125] Moreover, the aggregate accumulation of institutional knowledge may lead to a broad convergence of disclosure-related best practices, which could further reduce the costs of the required disclosures.

Overall, the market effects deriving from competition and innovation could enhance the efficiency and availability of relevant services, thereby lowering compliance costs. These positive externalities from standard reporting practices can provide additional market-wide cost savings to the extent that they reduce duplicative effort in the production and acquisition of information.[3126]

D. Other Economic Effects

The analysis of benefits and costs in section IV.C is generally based on the assumption that the final rules will not cause registrants to change how they manage climate-related risks, but rather how they produce the associated disclosures. In this section, we consider the

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3124 *See supra* note 1372 and accompanying text in section II.I.2.c.

3125 *See also supra* notes 2873, 3118 and accompanying text.

3126 *See* Christensen *et al.* (2021).

possibility that the rules may influence how some companies approach climate-related risks. For example, if agency conflicts currently prompt some managers to ignore long-run climate-related risks, in an effort to increase short-term cash flows, the additional transparency provided by the final rules may lead managers to focus more on long-run considerations if that is what their shareholders demand. Conversely, if some managers currently are over-prioritizing climate-related risks as compared to what investors view as optimal, the final rules may lead those managers to scale back their level of investment in managing climate-related risks. Generally, we expect that any resulting changes in behavior will primarily stem from investors' improved ability to assess managerial decisions. That is, to the extent the final rules prompt managers to alter their approach to climate-related risks, it may be because they expect that failing to do so might prompt a negative stock price reaction to the disclosures.[3127]

Registrants may change their behavior in response to the proposed disclosure requirements by managing exposures to certain physical or transition risks. For example, empirical evidence shows that mandatory reporting of GHG emissions results in reduced aggregate reported emissions among affected firms.[3128] The final rules will require the disclosure of the location of company properties or operations subject to material physical risks (Item 1502(a)(1)), which could allow investors to better assess companies' exposures to such risks. It is possible that, in response to or anticipation of investor reactions, companies may relocate properties or operations to geographical areas less exposed to physical risks or give

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3127 *See* M. Kahn, J. Matsusaka & C. Shu, *Divestment and Engagement: The Effect of Green Investors on Corporate Carbon Emissions* (Oct. 3, 2023), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4592023 (retrieved from SSRN Elsevier database).

3128 *See* Jeong-Bon Kim, *et al.*, *supra* note 2586; B. Downar, *et al.*, *supra* note 2776; S. Tomar, *Greenhouse Gas Disclosure and Emissions Benchmarking*, SMU Cox Sch. of Bus. Rsch. Paper No. 19-17 (2021), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3448904 (retrieved from SSRN Elsevier database); V. Jouvenot & P. Krueger, supra note 2775.

preference to such areas for future business activity. Any such changes to registrant behavior resulting from the final rules may come with the potential cost of lower productivity, profitability, or market share.[3129] In the case of relocation, for example, the alternate location may be more costly to operate. Similarly, we also recognize that some of the costs associated with the final rules may prompt some registrants to abandon or forgo adoption of material targets or goals relating to GHG emissions. To avoid direct costs of compliance or to simply report a lower emissions amount in their required disclosures, some registrants may take steps to reorganize their business in order to shift certain parts of their Scope 1 and Scope 2 emissions into the Scope 3 emissions category.[3130] This potential response from registrants obscures the registrants' true risk exposure and therefore could diminish the benefits of the disclosure related to investors' ability to assess exposure to climate-related transition risks.

Some commenters asserted that the compliance costs of the rules might cause some registrants to reduce their voluntary oversight of climate-related risks. For example, according to one commenter, devoting "resources to meeting the requirements of any final rules the Commission adopts… will detract from other climate-related reporting efforts."[3131] This commenter also asserted that the proposed requirement to "disclose internal information, such as internal carbon pricing, scenario planning, and related information if a company has an emission reduction target, could discourage companies from setting such targets."[3132] We recognize that

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3129 At the same time, we recognize that a registrant may optimize for both climate risks and productivity, as these factors are not necessarily mutually exclusive.

3130 *See* Lucas Mahieux, Haresh Sapra & Gaoqing Zhang, *Climate-Related Disclosures: What Are the Economic Trade-Offs?* (Dec. 1, 2023), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4507526 (retrieved from SSRN Elsevier database).

3131 *See* letter from API.

3132 *Id.*

some companies may pursue such avoidance strategies in response to the final rules. Other companies, however, may find the existence of disclosure requirements around climate-related targets and goals to be beneficial for signaling credible value-enhancing commitments to investors and hence may be motivated to engage in setting targets.[3133] More reliable and standardized disclosures about climate-related targets and goals will facilitate investors' understanding of the impact of those targets and goals, and hence could affect registrants' incentives for making such commitments, but the magnitude and direction of any such effects would depend upon registrants' decisions and investors' assessments about the value of those commitments rather than stemming directly from the final rules.

E. Effects on Efficiency, Competition, and Capital Formation

1. Efficiency

The final rules should have positive effects on market efficiency. As discussed above, the final rules should improve the informativeness and reliability of climate-related risks and financial disclosures. As a result of the disclosures required by the final rules, investors and other market participants should better understand the climate-related risks that registrants are facing, their potential impact (*e.g.*, on future cash flows), and registrants' ability to respond to and manage such risks. Investors and other market participants should thereby better evaluate registrants and make more informed investment and voting decisions. As a result, the required disclosures should reduce information asymmetry and mispricing in the market, improving

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[3133] Disclosures filed with the Commission are subject to greater liability and thus may be viewed as more credible than similar disclosures provided via other avenues (e.g., company sustainability reports). In addition, the final rules will require disclosure of details or specifics that some registrants may otherwise not provide in the absence of the final rules.

market efficiency. More efficient prices should improve capital formation by increasing overall public trust in markets, leading to greater investor participation and market liquidity.[3134]

Currently, investors may seek information on registrant's climate-related risks from various sources, including those outside of Commission filings. For example, the necessary information may only be available from company websites or from third-party service providers that collect information and offer their analysis for a fee. Once investors locate relevant disclosures, they may need to spend time organizing and compiling information in ways that facilitate comparisons across companies. Because the final rules will make the required disclosures available from a consistent source (i.e., Commission filings) and because the disclosures will be standardized and tagged, we expect the final rules to improve efficiency by reducing the costs associated with compiling and organizing information on climate-related risks and oversight.[3135]

We expect the climate-related disclosures mandated by the final rules will cause differential asset price and financing cost responses across companies and settings, as investors are more easily able to factor this information into their valuation decisions. These expected

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[3134] *See* Grewal, *et al.*, supra note 2653; M.E. Barth, *et al.*, *Textual Dimensions of Non-Financial Information, Stock Price Informativeness, and Proprietary Costs: Evidence from Integrated Reports*, (July 27, 2023), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3857927; see also D.S. Dhaliwal *et al.*, *Voluntary Nonfinancial Disclosure and the Cost of Equity Capital: The Initiation of Corporate Social Responsibility Reporting*, 86 Acct. Rev. 59 (2011); S. Kleimeier & M. Viehs, *Carbon Disclosure, Emission Levels, and the Cost of Debt*, (Jan. 7, 2018), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2719665 (retrieved from SSRN Elsevier database); E.M. Matsumura, *et al.*, *Climate Risk Materiality and Firm Risk*, supra note 2744. *But see* I. Goldstein & L. Yang, *Good Disclosure, Bad Disclosure*, 131 J. of Fin. Econ. 118 (2019).

[3135] One commenter stated that "[t]he Commission offers no support for the view that a rule aimed at consistency should be a stand-alone goal that will promote competition, efficiency, and capital formation." *See* Overdahl exhibit to letter from Chamber. To the extent that the commenter is asserting that the consistency achieved by the final rules does not promote or is somehow at odds with competition, efficiency, and capital formation, we disagree for the reasons outlined in this paragraph. Moreover, the Commission considers benefits and costs of the final rules in addition to the economic effects associated with efficiency, competition, and capital formation. *See* SEC Guidance on Economic Analysis (2012), *supra* note 2574.

improvements in market efficiency are broadly consistent with empirical research. For example, one academic study finds evidence that, among companies that voluntarily report emissions via the CDP questionnaire, those with higher emissions (relative to their size and industry peers) pay higher loan spreads.[3136] Another study examined more than 16,000 companies from 2016 through 2020 and found that investors were actively and directly pricing some transition risk into valuations, an action that resulted in a negative correlation between companies' CO_2 emissions and their price-to-earnings ratio.[3137]

Empirical research has also documented evidence of current market inefficiencies with respect to climate-related risks. For example, one study found that stock prices of food processing and agricultural companies may exhibit mispricing with respect to drought exposure.[3138] The study documented that drought-exposed companies report reduced future profitability, indicating that drought exposure is a financial risk.[3139] In an efficient market, this risk should result in trading activity that decreases the current stock price and increases the expected return (to compensate investors for bearing this risk). The study, however, found that drought-exposed companies deliver *lower* future returns relative to companies with less exposure, suggesting that the market initially under-reacts to drought exposure.[3140] In other

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3136 *See* S. Kleimeier & M. Viehs, *supra* note 3134.

3137 *See* Lazard Climate Center, *Inaugural Research Findings of the Lazard Climate Center* (Dec. 2021), available at https://www.lazard.com/research-insights/inaugural-research-findings-of-the-lazard-climate-center/; *see also* https://lazard.com/media/ge5oromo/lazard-climate-center-presentation-december-2021.pdf (presentation). The Lazard presentation notes, however, that the effects vary significantly across different types of GHG emissions, market capitalization, and sectors. Large capitalization companies (>$50 billion) experience greater valuation discounts, while larger emitters, such as energy companies, showed the most consistently negative correlation. On average, a 10% decrease in a large U.S. energy company's emissions corresponded with a 3.9% increase in its price-to-earnings ratio.

3138 *See* H. Hong, *et al.*, *supra* note 2739.

3139 *See id.*

3140 *See id.*

words, the market fails to sufficiently incorporate the risk of drought exposure into the current stock price, resulting in investors holding mispriced assets and bearing risk for which they are not appropriately compensated. Consistent with this finding, survey responses from institutional investors indicated that such investors believed that equity valuations do not fully reflect climate-related risks.[3141] The final rules may help address these market inefficiencies by eliciting more consistent and reliable information about climate-related risks so that those risks can be better incorporated into asset prices.

We also expect the final rules to increase efficiency by improving comparability of climate-related disclosures and requiring them to be filed in a machine-readable data language (i.e., Inline XBRL).[3142] As discussed in section IV.C.2.i, efficiency gains from standardized reporting practices can provide market-wide cost savings to registrants in the long-term, to the extent that they reduce duplicative effort in registrants' production and acquisition of information (e.g., certain data or third-party services related to preparing the required disclosures, including the reporting of emissions data, may become cheaper in the long run as heightened demand spurs competition, innovation, and economies of scale). Finally, more standardized reporting should also reduce investors' costs of acquiring and processing climate-related information by facilitating investors' analysis of a registrant's disclosure and assessing its management of climate-related risks against those of its competitors.

The inclusion of climate-related information in Commission filings using a machine-readable data language (i.e., Inline XBRL), rather than external reports or company websites,

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3141 *See* Krueger, *et al.*, *supra* note 2790.

3142 *See* letters from Impact Capital Managers (indicating that the Inline XBRL requirement will contribute toward the goal of eliciting more consistent, comparable, and reliable disclosure); and Climate Advisers (stating that tagging the new disclosures in Inline XBRL should, by allowing the disclosed information to be more readily incorporated into investors' analyses, promote the efficiency of the U.S. capital markets).

should also make it easier for investors to find and compare this information. In that regard, XBRL requirements have been observed to reduce the informational advantages of informed traders and lead to lower cost of capital and higher stock liquidity for filers that provide tagged disclosures.[3143]

We acknowledge commenters who stated that proposed amendments could decrease efficiency by reducing the incentives for reporting companies to develop business strategies, transition plans, or goals, because the amendments would require disclosure of these strategies, plans or goals.[3144] According to these commenters, the benefits of developing these elements could be outweighed by the direct and indirect costs of disclosing them. While this may occur in some circumstances, the efficiency loss is expected to be relatively low as the required disclosures are not highly granular. Thus, in many cases, we believe the benefits of developing business strategies, transition plans or goals will exceed the costs of such disclosure. But we recognize that, more generally, the final rules may divert some resources away from what their best use would otherwise be. As explained above, by removing some of the more prescriptive elements of the proposed rules that could require disclosure of a registrant's competitively sensitive information, the final rules mitigate this concern.

[3143] *See, e.g.*, N. Bhattacharya, Y.J. Cho & J.B. Kim, *Leveling the Playing Field Between Large and Small Institutions: Evidence from the SEC's XBRL Mandate*, 93 Acct. Rev. 51 (2018); B. Li, Z. Liu, W. Qiang & B. Zhang, *The Impact of XBRL Adoption on Local Bias: Evidence from Mandated U.S. Filers*, 39 J. of Acct. and Pub. Policy (2020); W. Sassi, H. Ben Othman & K. Hussainey, *The Impact of Mandatory Adoption of XBRL on Firm's Stock Liquidity: A Cross-Country Study*, 19 J. of Fin. Rep. and Acct. 299 (2021); C. Ra & H. Lee, *XBRL Adoption, Information Asymmetry, Cost of Capital, and Reporting Lags*, 10 iBusiness 93 (2018); S.C. Lai, Y.S. Lin, Y.H. Lin & H.W. Huang, *XBRL Adoption and Cost of Debt*, Int'l. J. of Acct. & Info. Mgmt. (2015); Cong *et al.*, supra note 2948.

[3144] *See, e.g.*, letter from Cato Inst.; Overdahl exhibit to letter from Chamber; and Motor & Equipment Manufacturers Association.